

CYCLIC AUTOCORRELATION, PERMUTATION FIBERS, AND A LAYERED SYMPLECTIC MODEL FOR PITCH-CLASS SPACE

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ABSTRACT. We construct a layered model that separates pitch-class content from the order in which its notes are traversed. For $n \geq 2$ and $1 \leq k \leq n$, we consider k -note patterns of distinct pitch classes in $G_n = \mathbb{Z}/n\mathbb{Z}$. Their directed cyclic gaps recover the traversal once a root is specified, while the full complex Fourier transform of the order signal recovers the rooted pattern directly. Content autocorrelation instead summarizes interval content: it is equivalent to Fourier power but need not determine the pitch-class set. We prove the relevant finite Wiener–Khinchin identities and characterize gap periodicity by support in an annihilator subgroup.

The full content and order coordinates give an injective map into $\mathbb{C}^{n-1} \times \mathbb{C}^k$. We equip this ambient space with a chosen weighted standard exact symplectic form, construct Hamiltonian circle extensions of the finite musical actions, and distinguish finite root-forgetting from circle reduction. On the free transposition locus, reduction and the residual cyclic quotient produce symplectic manifolds and orbifold strata; inversion descends anti-symplectically. Cardinality indexes a disjoint family of these spaces. A companion Lean library verifies the discrete foundation and the 223-class nonempty twelve-tone census; the higher geometry is established here by paper proofs. Finite visualization estimates explain when rendered fibers remain separated, without asserting that the supplied plots preserve every metric or topological property.

1. INTRODUCTION AND STATEMENT OF RESULTS

Let $n \geq 2$ and write $G_n = \mathbb{Z}/n\mathbb{Z}$. A nonempty subset $S \subset G_n$ records pitch-class content; an ordered k -tuple of distinct classes

$$p = (p_0, \dots, p_{k-1}) \in \mathcal{P}_{n,k}$$

records a rooted cyclic traversal of that content. The guiding principle of this paper is that content and order must be separated before one attempts a continuous geometry. The model proposed here is not a single manifold

Date: September 13, 2026 (mathematical and expository review).

2020 *Mathematics Subject Classification.* Primary 53D20, 42A16; Secondary 05E18, 57S17, 00A65.

Key words and phrases. cyclic autocorrelation, discrete Fourier analysis, pitch-class sets, permutation fibers, symplectic reduction, orbifold strata, twelve-tone systems.

carrying all cardinalities at once, but a cardinality-graded family of Fourier-coordinate thickenings.

The content layer is controlled by the cyclic autocorrelation of the indicator of S , following the combinatorial viewpoint of Duncan [5] and the Fourier-analytic perspective developed by Quinn [8], Amiot [9], and Yust [10]. The order layer is controlled by two signals: the gap signal, which detects periodic interval cycles in the sense suggested by Slonimsky, and the order signal, which remembers the full rooted cyclic arrangement. The resulting spectral coordinates fit naturally into the ambient complex vector space

$$\mathcal{M}_{n,k} = \mathbb{C}^{n-1} \times \mathbb{C}^k,$$

and it is this chosen ambient space that carries the symplectic structure defined below.

Objects and scope. For example, $(0, 4, 7)$ and $(0, 7, 4)$ traverse the same content $\{0, 4, 7\}$ but have different directed cyclic gaps, $(4, 3, 5)$ and $(7, 9, 8)$. Their content descriptors agree; their full order spectra distinguish them. Throughout, a root means the chosen first pitch of a cyclic traversal, not a harmonic root inferred from tonal function. The model records pitch classes and their cyclic order, with register, timing, dynamics, and repeated-note events omitted. In particular, closing a finite row segment into a cycle is a modeling choice.

The distinction between a coordinate and an invariant is central. The full complex content transform, with cardinality restoring its omitted zeroth coefficient, determines the content set. Its squared magnitudes determine autocorrelation and can identify different content sets. The full complex order transform determines the rooted pattern; its magnitudes need not. The auxiliary gap transform diagnoses interval-cycle periodicity and is not an additional factor of the symplectic embedding.

Verification status. The companion Lean 4 library implements the discrete contracts F01–F10: reconstruction, ordering and cyclic-fiber counts, finite complex Wiener–Khinchin, content invariance, full order-spectrum reconstruction, layered injectivity, the twelve-tone T/I census, and exact representation-boundary examples. Its release manifest is `formal/PROOFS.md`; `formal/TRUST.md` records the pinned dependencies and axiom audit. General periodicity/support and all differential and symplectic results remain paper proofs. The formal library does not certify the numerical visualizer, historical labels, or musical utility.

The main results are as follows.

Theorem A. For every content set $S \subset G_n$, the cyclic autocorrelation of its indicator function is the inverse Fourier transform of $|\widehat{\chi_S}|^2$. For every ordered pattern $p \in \mathcal{P}_{n,k}$, the cyclic autocorrelation of its order signal is the inverse Fourier transform of the squared moduli of its order spectrum. Consequently the squared magnitudes of the content spectrum, and hence the content autocorrelation, are invariant under transposition and inversion; see [theorems 3.2](#) and [5.3](#) and [corollary 3.3](#).

Theorem B. For $d \mid k$, a gap signal satisfies $g(m + d) = g(m)$ for all m if and only if its Fourier transform is supported on the multiples of k/d . In particular, equal-step cycles have period 1 and DC-only gap spectrum; d -periodic interpolations have support on the d -point harmonic lattice; see [theorem 4.1](#) and [corollary 4.2](#).

Theorem C. For every $1 \leq k \leq n$ there is an injective layered Fourier embedding

$$\iota_{n,k} : \mathcal{P}_{n,k} \hookrightarrow \mathcal{M}_{n,k} = \mathbb{C}^{n-1} \times \mathbb{C}^k$$

whose first factor records content and whose second factor records rooted cyclic order. The form

$$\omega_{n,k}^{(\lambda)} = \frac{i}{2} \sum_{j=1}^{n-1} dX_j \wedge d\overline{X_j} + \lambda \frac{i}{2} \sum_{\ell=0}^{k-1} dY_\ell \wedge d\overline{Y_\ell}$$

is exact symplectic on $\mathcal{M}_{n,k}$ and becomes the full action-angle form

$$\sum_{j=1}^{n-1} dI_j \wedge d\theta_j + \lambda \sum_{\ell=0}^{k-1} dJ_\ell \wedge d\phi_\ell$$

on the nonvanishing-mode locus; see [proposition 6.2](#) and [theorems 7.2](#) and [7.4](#).

Theorem D. Pitch transposition and cyclic reindexing admit commuting Hamiltonian circle extensions on $\mathcal{M}_{n,k}$ with explicit moment maps, while inversion extends to an anti-symplectic involution. The discrete musical symmetries are recovered by restricting these continuous actions to the finite subgroups coming from the roots of unity; see [theorems 8.2, 8.4](#) and [8.6](#).

Theorem E. The total space

$$\mathcal{M}_n = \bigsqcup_{k=1}^n \mathcal{M}_{n,k}$$

has a cardinality grading with the disjoint-union topology and no cross-component attachments. On the open locus where the transposition action is free and at regular moment values, Marsden–Weinstein reduction produces a symplectic manifold; quotienting by the finite cyclic reindexing group then yields a symplectic orbifold. The singular orbit-type strata are determined by explicit stabilizers, analyzed in [appendix A](#); see also [proposition 9.3](#).

The proof strategy is constructive. Sections 2–5 build the discrete seed and its Fourier invariants. Sections 6 and 7 construct the layered embedding and symplectic thickening. Sections 8 and 9 analyze the continuous symmetries, reductions, and cardinality stratification. Appendix A isolates the stabilizer calculations and the reduced-orbifold local models. Two short appendices record the exact coordinate-extraction algorithm and the role of deliberate redundancy in the layered embedding.

2. DISCRETE SEED: CONTENT, ORDER, AND QUOTIENT ACTIONS

2.1. Pitch classes and content. Fix $n \geq 2$ and write $G_n = \mathbb{Z}/n\mathbb{Z}$. We denote by

$$T_t(x) = x + t, \quad I(x) = -x, \quad x, t \in G_n$$

the transposition and inversion actions. These transformations define an action of the abstract dihedral group

$$D_n = G_n \rtimes \{\pm 1\},$$

acting on G_n by affine maps $x \mapsto \pm x + t$. The action is faithful for $n \geq 3$; at $n = 2$, inversion acts trivially, so the action has a kernel.

Definition 2.1 (Pitch-class content). A *content set* is a nonempty subset $S \subset G_n$. Its *cardinality* is $|S| = k$. The content classes modulo transposition and modulo transposition/inversion are the orbit sets

$$\mathcal{C}_n^T = (2^{G_n} \setminus \{\emptyset\})/G_n, \quad \mathcal{C}_n^{TI} = (2^{G_n} \setminus \{\emptyset\})/D_n.$$

When $n = 12$, the orbit count is easy to compute directly by Burnside's lemma or brute force. There are 224 T/I -classes including the empty set and 223 nonempty T/I -classes.

Remark 2.2 (Forte classification and harmonic nomenclature). The standard nomenclature for these classes is due to Forte [6], who assigns each T/I -class a canonical label of the form k - n , where k is the cardinality and n is the ordinal within the cardinality group. Under this system, many of the 223 classes correspond to objects familiar to every musician: 3-11 is the major/minor triad, 4-27 is the dominant/half-diminished seventh chord, 7-35 is the diatonic scale, 4-28 is the diminished seventh chord, and 6-35 is the whole-tone collection. Others—such as 5-Z12, 6-Z44, or 8-22—lack familiar common names but occupy equally definite positions in the pitch-class universe. The prefix “Z” in Forte’s notation signals that a set class belongs to a homometric pair: 5-Z12 and 5-Z36, for instance, share the same interval content yet are not T/I -equivalent (see the remark on homometry in §3 below). Common names are selected annotations; their absence is not evidence of historical non-use. Catalog labels and comprehensive historical coverage are not certified here.

2.2. Ordered patterns and cyclic fibers.

Definition 2.3 (Ordered pattern). For $1 \leq k \leq n$, let

$$\mathcal{P}_{n,k} = \left\{ p = (p_0, \dots, p_{k-1}) \in G_n^k : p_i \neq p_j \text{ for } i \neq j \right\}$$

be the set of ordered k -tuples of distinct pitch classes. The *content projection* is

$$c : \mathcal{P}_{n,k} \rightarrow 2^{G_n}, \quad c(p) = \{p_0, \dots, p_{k-1}\}.$$

Fix a content set S of size k . Then

$$\text{Ord}(S) = \{p \in \mathcal{P}_{n,k} : c(p) = S\}$$

is the set of all orderings of S , hence $|\text{Ord}(S)| = k!$.

There are two equivalent ways to remove the choice of starting point. The first takes a quotient by cyclic reindexing:

$$\sigma \cdot (p_0, \dots, p_{k-1}) = (p_1, \dots, p_{k-1}, p_0),$$

which defines an action of the cyclic group C_k on $\text{Ord}(S)$. The quotient $\text{Ord}(S)/C_k$ records the cyclic order while forgetting the choice of starting point. The second selects a representative by fixing $r \in S$ and considering only tuples with first entry r . This selection is not a second group quotient.

Definition 2.4 (Rooted cyclic fiber). Let $S \subset G_n$ with $|S| = k$ and let $r \in S$. The *rooted cyclic fiber at r* is

$$\text{Cyc}_r(S) = \{p \in \text{Ord}(S) : p_0 = r\}.$$

Proposition 2.5 (Size of the rooted cyclic fiber). *If $|S| = k$ and $r \in S$, then*

$$|\text{Cyc}_r(S)| = (k-1)!.$$

Equivalently, the quotient $\text{Ord}(S)/C_k$ has $(k-1)!$ elements.

Proof. Fix $p_0 = r$. The remaining $k-1$ entries may be arranged arbitrarily as a permutation of $S \setminus \{r\}$. This yields $(k-1)!$ possibilities. Since every cyclic orbit in $\text{Ord}(S)$ contains exactly one representative with first entry equal to r , the same number counts $\text{Ord}(S)/C_k$. $\square$

The full fiber of $c : \mathcal{P}_{n,k} \rightarrow 2^{G_n}$ over S has size $k!$. After fixing a root r , or equivalently after quotienting by cyclic reindexing, the cyclic fiber has size $(k-1)!$.

2.3. Gap words and reconstruction.

Definition 2.6 (Gap word). For $p = (p_0, \dots, p_{k-1}) \in \mathcal{P}_{n,k}$ define its *gap word*

$$\delta(p) = (\delta_0, \dots, \delta_{k-1}) \in G_n^k, \quad \delta_i = p_{i+1} - p_i,$$

where indices are taken modulo k .

The gap word records the successive directed intervals of the cyclic traversal. It always satisfies

$$\sum_{i=0}^{k-1} \delta_i = 0 \quad \text{in } G_n.$$

Proposition 2.7 (Reconstruction from root and gap word). *Let $r \in G_n$ and let $\delta = (\delta_0, \dots, \delta_{k-1}) \in G_n^k$. Define*

$$p_0 = r, \quad p_m = r + \sum_{i=0}^{m-1} \delta_i \quad (1 \leq m \leq k-1).$$

Then $p = (p_0, \dots, p_{k-1})$ lies in $\mathcal{P}_{n,k}$ and has cyclic gap word δ if and only if

- (a) $\sum_{i=0}^{k-1} \delta_i = 0$ in G_n , and
 (b) the partial sums $0, \delta_0, \delta_0 + \delta_1, \dots, \sum_{i=0}^{k-2} \delta_i$ are pairwise distinct in G_n .
 In that case p is uniquely determined by (r, δ) .

Proof. By construction,

$$p_{m+1} - p_m = \delta_m, \quad 0 \leq m \leq k-2.$$

Moreover

$$p_0 - p_{k-1} = - \sum_{i=0}^{k-2} \delta_i = \delta_{k-1}$$

holds precisely when $\sum_i \delta_i = 0$. Distinctness of the entries is equivalent to distinctness of the partial sums. Uniqueness is immediate from the recursive formula. $\square$

This elementary proposition is the first formal bridge between permutation space and interval cycles: the cyclic order is encoded by a constrained word in the group G_n , and Slonimsky-style cycle families appear as special periodic classes of those words.

3. CONTENT LAYER: CYCLIC AUTOCORRELATION AND SPECTRAL INVARIANTS

3.1. Finite Fourier transform on G_n . Let $\eta_n = e^{2\pi i/n}$. For a function $f : G_n \rightarrow \mathbb{C}$, define its discrete Fourier transform by

$$\widehat{f}(j) = \sum_{x \in G_n} f(x) \eta_n^{-jx}, \quad j \in G_n.$$

The inverse formula is

$$f(x) = \frac{1}{n} \sum_{j \in G_n} \widehat{f}(j) \eta_n^{jx}.$$

For a content set $S \subset G_n$, let χ_S be its indicator function:

$$\chi_S(x) = \begin{cases} 1, & x \in S, \\ 0, & x \notin S. \end{cases}$$

3.2. Cyclic autocorrelation.

Definition 3.1 (Content autocorrelation). The *cyclic autocorrelation* of S is the function

$$A_S(\tau) = \sum_{x \in G_n} \chi_S(x) \chi_S(x + \tau), \quad \tau \in G_n.$$

Thus $A_S(\tau)$ counts the number of ordered pairs $(x, y) \in S \times S$ with $y - x = \tau$. In particular $A_S(0) = |S|$ and $A_S(-\tau) = A_S(\tau)$.

Theorem 3.2 (Finite Wiener–Khinchin for content). *For every content set $S \subset G_n$,*

$$\widehat{A}_S(j) = |\widehat{\chi}_S(j)|^2 \quad (j \in G_n).$$

Equivalently,

$$A_S(\tau) = \frac{1}{n} \sum_{j \in G_n} |\widehat{\chi}_S(j)|^2 \eta_n^{j\tau}.$$

Proof. We compute directly:

$$\begin{aligned} \widehat{A}_S(j) &= \sum_{\tau \in G_n} A_S(\tau) \eta_n^{-j\tau} \\ &= \sum_{\tau \in G_n} \sum_{x \in G_n} \chi_S(x) \chi_S(x + \tau) \eta_n^{-j\tau}. \end{aligned}$$

Set $y = x + \tau$, so $\tau = y - x$. Then

$$\begin{aligned} \widehat{A}_S(j) &= \sum_{x, y \in G_n} \chi_S(x) \chi_S(y) \eta_n^{-j(y-x)} \\ &= \left(\sum_{y \in G_n} \chi_S(y) \eta_n^{-jy} \right) \left(\sum_{x \in G_n} \chi_S(x) \eta_n^{jx} \right) \\ &= \widehat{\chi}_S(j) \overline{\widehat{\chi}_S(j)} = |\widehat{\chi}_S(j)|^2. \end{aligned}$$

The inverse formula gives the second identity. $\square$

Corollary 3.3 (Transposition and inversion invariance). *If $S' = T_t(S)$ is a transposition of S , then*

$$\widehat{\chi}_{S'}(j) = \eta_n^{-jt} \widehat{\chi}_S(j) \quad \text{and hence} \quad A_{S'} = A_S.$$

If $S' = I(S)$ is the inversion of S , then

$$\widehat{\chi}_{S'}(j) = \widehat{\chi}_S(-j) \quad \text{and hence} \quad A_{S'} = A_S.$$

Therefore A_S , equivalently $|\widehat{\chi}_S(j)|^2$, is a T/I -invariant of content.

Proof. For transposition,

$$\chi_{S'}(x) = \chi_S(x - t),$$

so

$$\widehat{\chi}_{S'}(j) = \sum_x \chi_S(x - t) \eta_n^{-jx} = \eta_n^{-jt} \sum_u \chi_S(u) \eta_n^{-ju}.$$

For inversion,

$$\chi_{S'}(x) = \chi_S(-x),$$

so

$$\widehat{\chi}_{S'}(j) = \sum_x \chi_S(-x) \eta_n^{-jx} = \sum_u \chi_S(u) \eta_n^{ju} = \widehat{\chi}_S(-j).$$

Taking moduli squared and invoking [theorem 3.2](#) proves the claim. $\square$

Remark 3.4 (Homometry). Two content sets $S, T \subset G_n$ are homometric if $A_S = A_T$, equivalently if

$$|\widehat{\chi_S}(j)| = |\widehat{\chi_T}(j)| \quad \text{for all } j \in G_n.$$

Thus cyclic autocorrelation is complete for interval content but not, in general, for content itself. Homometric sets that are not T/I -equivalent are $\mathbb{Z}$ -related in pitch-class set theory (see [6], [12]). Yust and Amiot [11] analyze the non-spectral component of transposition-invariant information, offering a complementary perspective on the information lost by the magnitude-only description.

3.3. Moment coordinates. The previous theorem shows that autocorrelation is a function of the nonnegative numbers

$$I_j = \frac{1}{2} |\widehat{\chi_S}(j)|^2, \quad j = 1, \dots, n-1.$$

These will reappear as action variables in the symplectic thickening. At this stage it is enough to note that the content autocorrelation is an affine linear image of the vector $(I_1, \dots, I_{n-1})$, with $|S| = \widehat{\chi_S}(0)$ serving as an additional cardinality parameter.

4. ORDER LAYER I: GAP WORDS AND SLONIMSKY SPECTRAL SPARSITY

4.1. The gap signal. The gap word records directed steps in G_n . To convert it into a Fourier-compatible signal, define

$$g_p(m) = \eta_n^{\delta_m(p)} \in S^1, \quad m \in \mathbb{Z}/k\mathbb{Z}.$$

Its Fourier transform on $\mathbb{Z}/k\mathbb{Z}$ is

$$G_p(\ell) = \sum_{m=0}^{k-1} g_p(m) \zeta_k^{-\ell m}, \quad \zeta_k = e^{2\pi i/k}, \quad 0 \leq \ell \leq k-1.$$

The signal g_p does not remember the absolute root p_0 , but it encodes precisely the directed interval-cycle structure of the pattern. For the Slonimsky viewpoint this is the essential order observable.

4.2. Periodicity versus Fourier support. The next theorem relates a specified shift period to a specified subgroup of Fourier modes. A proper period forces structured spectral sparsity; having few nonzero coefficients by itself does not imply a proper period.

Theorem 4.1 (Shift-periodic signals and sparse spectra). *Let d be a positive divisor of k and let $g : \mathbb{Z}/k\mathbb{Z} \rightarrow \mathbb{C}$. Then the following are equivalent:*

- (i) $g(m+d) = g(m)$ for all m ;
- (ii) $\widehat{g}(\ell) = 0$ whenever ℓ is not a multiple of k/d .

Proof. Let $T_d g(m) = g(m+d)$. By a standard Fourier calculation,

$$\widehat{T_d g}(\ell) = \zeta_k^{\ell d} \widehat{g}(\ell).$$

Hence $T_d g = g$ if and only if

$$(\zeta_k^{\ell d} - 1)\widehat{g}(\ell) = 0 \quad \text{for all } \ell.$$

Now $\zeta_k^{\ell d} = 1$ if and only if $k \mid \ell d$, equivalently if and only if ℓ is a multiple of k/d . Therefore $T_d g = g$ exactly when $\widehat{g}(\ell)$ vanishes outside those multiples. $\square$

Corollary 4.2 (Slonimsky interval cycles). *For a positive divisor d of k , if the gap signal g_p has shift period d , then the gap spectrum G_p is supported on the harmonic lattice*

$$\{0, k/d, 2k/d, \dots, (d-1)k/d\} \subset \mathbb{Z}/k\mathbb{Z}.$$

In particular, equal-step cycles correspond to DC-only gap spectra, while d -periodic interval patterns correspond to support contained in that d -point lattice.

Remark 4.3. The theorem supplies a way to analyze interval-cycle families associated with Slonimsky. Identifying a particular catalogued interpolation requires checking its actual gap word; no complete classification of the catalogue by Fourier support is asserted. Nor does arbitrary sparsity suffice: the valid pattern $(0, 1, 3, 2, 7, 5)$ in G_8 has gaps $(1, 2, 7, 5, 6, 3)$ and satisfies $g(m+3) = -g(m)$. Its even Fourier modes vanish, yet its gap word has no proper period. The surviving modes lie in a coset rather than the required annihilator subgroup.

4.3. Gap autocorrelation. Define the gap autocorrelation by

$$B_p(r) = \sum_{m=0}^{k-1} \overline{g_p(m)} g_p(m+r), \quad r \in \mathbb{Z}/k\mathbb{Z}.$$

An explicit calculation confirms the finite Wiener–Khinchin identity:

$$\begin{aligned} \widehat{B}_p(\ell) &= \sum_{r,m} \overline{g_p(m)} g_p(m+r) \zeta_k^{-\ell r} \\ &= \overline{G_p(\ell)} G_p(\ell) = |G_p(\ell)|^2. \end{aligned}$$

The intermediate step follows by substituting $u = m + r$ and separating the double sum. Thus gap autocorrelation is determined by $|G_p(\ell)|^2$, just as content autocorrelation is determined by $|\widehat{\chi_S}(j)|^2$. In both cases taking magnitudes discards phase information; neither autocorrelation is asserted to reconstruct its underlying signal. The G_p -coordinates are used here as auxiliary interval-cycle diagnostics. They are not part of the symplectic embedding $\iota_{n,k}$ below unless one enlarges $\mathcal{M}_{n,k}$ by an additional G -factor.

Remark 4.4 (Step-histogram diversity). Put $h_p(s) = \#\{i : \delta_i = s\}$ and

$$H(p) = - \sum_{s \in G_n} \frac{h_p(s)}{k} \log_2 \frac{h_p(s)}{k}, \quad 0 \log_2 0 = 0.$$

Then $0 \leq H(p) \leq \log_2 \min(n, k)$, and $H(p) = 0$ exactly when all gaps coincide. Transposition and cyclic reindexing preserve the histogram, while inversion permutes its bins, so all preserve entropy. This measures step diversity, not smoothness or reading order: the full twelve-note cycles $p_m = m$ and $q_m = 5m$ ($0 \leq m < 12$) in G_{12} both have entropy zero but circular step lengths 1 and 5. A separate smoothness cost could be $k^{-1} \sum_m \min(\delta_m, n - \delta_m)$ using integer representatives.

5. ORDER LAYER II: FAITHFUL ORDER SIGNAL AND EXACT RECONSTRUCTION

The gap signal is ideal for cycle periodicity, but it forgets the root. To obtain a faithful ordered encoding we use the absolute order signal.

5.1. Order signal.

Definition 5.1 (Order signal). For $p = (p_0, \dots, p_{k-1}) \in \mathcal{P}_{n,k}$ define

$$s_p(m) = \eta_n^{p_m} \in S^1, \quad m \in \mathbb{Z}/k\mathbb{Z}.$$

Its Fourier transform is

$$Y_p(\ell) = \sum_{m=0}^{k-1} s_p(m) \zeta_k^{-\ell m}, \quad 0 \leq \ell \leq k-1.$$

The sequence s_p remembers the ordered pitch classes themselves, not only their successive differences.

Proposition 5.2 (Exact reconstruction from order Fourier data). *For fixed n and k , the map*

$$p \mapsto (Y_p(0), \dots, Y_p(k-1)) \in \mathbb{C}^k$$

is injective on $\mathcal{P}_{n,k}$.

Proof. By inverse Fourier transform on $\mathbb{Z}/k\mathbb{Z}$,

$$s_p(m) = \frac{1}{k} \sum_{\ell=0}^{k-1} Y_p(\ell) \zeta_k^{\ell m}.$$

Hence the vector $(Y_p(\ell))_{\ell=0}^{k-1}$ uniquely determines the sequence $(s_p(m))_{m=0}^{k-1}$. Each $s_p(m)$ is an n th root of unity, and therefore uniquely determines $p_m \in G_n$. Thus the whole ordered tuple p is recovered. $\square$

5.2. Order autocorrelation. Define the order autocorrelation

$$C_p(r) = \sum_{m=0}^{k-1} \overline{s_p(m)} s_p(m+r), \quad r \in \mathbb{Z}/k\mathbb{Z}.$$

Theorem 5.3 (Finite Wiener–Khinchin for order). *For every $p \in \mathcal{P}_{n,k}$,*

$$\widehat{C}_p(\ell) = |Y_p(\ell)|^2 \quad (0 \leq \ell \leq k-1).$$

Proof. Expanding directly:

$$\begin{aligned}\widehat{C}_p(\ell) &= \sum_{r,m} \overline{s_p(m)} s_p(m+r) \zeta_k^{-\ell r} \\ &= \overline{Y_p(\ell)} Y_p(\ell) = |Y_p(\ell)|^2.\end{aligned}$$

The intermediate step follows by substituting $u = m + r$ and separating the double sum, as in [theorem 3.2](#). $\square$

Thus the order layer has its own nonnegative observables

$$J_\ell = \frac{1}{2} |Y_p(\ell)|^2, \quad 0 \leq \ell \leq k-1.$$

The J_ℓ are invariant under pitch transposition and cyclic reindexing. Inversion permutes them by $\ell \mapsto -\ell$; it need not preserve each indexed value.

5.3. Symmetries of the order signal. Two order symmetries are particularly natural.

Proposition 5.4 (Cyclic reindexing). *Let σ be the cyclic reindexing operator*

$$\sigma \cdot (p_0, \dots, p_{k-1}) = (p_1, \dots, p_{k-1}, p_0).$$

Then

$$Y_{\sigma^r p}(\ell) = \zeta_k^{\ell r} Y_p(\ell).$$

Proof. The order signal of $\sigma^r p$ is $s_p(m+r)$. Therefore

$$\begin{aligned}Y_{\sigma^r p}(\ell) &= \sum_{m=0}^{k-1} s_p(m+r) \zeta_k^{-\ell m} \\ &= \zeta_k^{\ell r} \sum_{u=0}^{k-1} s_p(u) \zeta_k^{-\ell u} = \zeta_k^{\ell r} Y_p(\ell).\end{aligned}$$

$\square$

Proposition 5.5 (Pitch transposition and inversion). *Let $T_t p = (p_0 + t, \dots, p_{k-1} + t)$ and $I p = (-p_0, \dots, -p_{k-1})$. Then*

$$Y_{T_t p}(\ell) = \eta_n^t Y_p(\ell), \quad Y_{I p}(\ell) = \overline{Y_p(-\ell)}.$$

Proof. Transposition multiplies every term of s_p by η_n^t , so the same scalar factors out of the Fourier transform. Inversion sends $s_p(m)$ to $\overline{s_p(m)}$, hence

$$Y_{I p}(\ell) = \sum_m \overline{s_p(m)} \zeta_k^{-\ell m} = \sum_m s_p(m) \zeta_k^{\ell m} = \overline{Y_p(-\ell)}.$$

$\square$

6. THE LAYERED FOURIER EMBEDDING

We now assemble the content and order observables.

6.1. Content coordinates. For $p \in \mathcal{P}_{n,k}$ let $S = c(p)$ and define

$$X_p(j) = \widehat{\chi_S}(j), \quad 1 \leq j \leq n-1.$$

We omit the zeroth mode because $X_p(0) = |S| = k$ is constant on the cardinality- k stratum.

6.2. The total embedding.

Definition 6.1 (Layered Fourier embedding). For fixed (n, k) define

$$\iota_{n,k} : \mathcal{P}_{n,k} \rightarrow \mathbb{C}^{n-1} \times \mathbb{C}^k, \quad \iota_{n,k}(p) = ((X_p(j))_{j=1}^{n-1}, (Y_p(\ell))_{\ell=0}^{k-1}).$$

Proposition 6.2 (Injectivity of the layered embedding). *The map $\iota_{n,k}$ is injective.*

Proof. By [proposition 5.2](#), the order Fourier coordinates $(Y_p(\ell))_{\ell=0}^{k-1}$ alone determine p . Therefore the larger map $\iota_{n,k}$ is injective as well. $\square$

The significance of $\iota_{n,k}$ is conceptual rather than minimal: the X -coordinates isolate content geometry, while the Y -coordinates isolate order geometry. Their coexistence makes the layered structure visible. The embedding is defined before quotienting by transposition, inversion, or cyclic reindexing. Quotient-level visualizations later in the paper require either choosing representatives or passing to invariant coordinates such as autocorrelation or magnitude data.

7. SYMPLECTIC THICKENING

7.1. Ambient complex manifold. Here “thickening” means placing the finite embedded set inside a specified ambient symplectic manifold. It does not equip the finite set with positive-dimensional tangent directions. The full ambient coordinates are independent, whereas musical seed points satisfy relations such as $X_{n-j} = \overline{X_j}$ and $\sum_{\ell} |Y_{\ell}|^2 = k^2$. Generic ambient points need not represent pitch-class patterns. The product form and its positive layer weight are choices of model, motivated by the separated coordinates rather than forced by injectivity alone.

Fix $\lambda > 0$. For each k set

$$\mathcal{M}_{n,k} = \mathbb{C}^{n-1} \times \mathbb{C}^k$$

with coordinates

$$(X_1, \dots, X_{n-1}, Y_0, \dots, Y_{k-1}).$$

Definition 7.1 (Layered symplectic form). Define the 2-form

$$\omega_{n,k}^{(\lambda)} = \frac{i}{2} \sum_{j=1}^{n-1} dX_j \wedge d\overline{X_j} + \lambda \frac{i}{2} \sum_{\ell=0}^{k-1} dY_{\ell} \wedge d\overline{Y_{\ell}}. \quad (7.1)$$

In real coordinates $X_j = x_j + iy_j$ and $Y_{\ell} = u_{\ell} + iv_{\ell}$,

$$\omega_{n,k}^{(\lambda)} = \sum_{j=1}^{n-1} dx_j \wedge dy_j + \lambda \sum_{\ell=0}^{k-1} du_{\ell} \wedge dv_{\ell}.$$

Theorem 7.2 (Symplectic thickening). *For every n, k and $\lambda > 0$, the pair $(\mathcal{M}_{n,k}, \omega_{n,k}^{(\lambda)})$ is an exact symplectic manifold, and*

$$\iota_{n,k}(\mathcal{P}_{n,k}) \subset \mathcal{M}_{n,k}$$

is an injective discrete subset. Hence $(\mathcal{M}_{n,k}, \omega_{n,k}^{(\lambda)})$ is a symplectic thickening of the discrete pattern space $\mathcal{P}_{n,k}$.

Proof. The form (7.1) is the direct sum of standard symplectic forms on complex vector spaces, hence closed and nondegenerate. It is exact with primitive

$$\alpha_{n,k}^{(\lambda)} = \frac{i}{4} \sum_{j=1}^{n-1} (X_j d\overline{X}_j - \overline{X}_j dX_j) + \lambda \frac{i}{4} \sum_{\ell=0}^{k-1} (Y_\ell d\overline{Y}_\ell - \overline{Y}_\ell dY_\ell),$$

since $d\alpha_{n,k}^{(\lambda)} = \omega_{n,k}^{(\lambda)}$. Injectivity of the discrete embedding was proved in [proposition 6.2](#). $\square$

Proposition 7.3 (Ambient rescaling of the weight parameter). *For $\lambda > 0$, the linear map*

$$F_\lambda : \mathcal{M}_{n,k} \rightarrow \mathcal{M}_{n,k}, \quad F_\lambda(X, Y) = (X, \sqrt{\lambda}Y)$$

is a symplectomorphism from $(\mathcal{M}_{n,k}, \omega_{n,k}^{(\lambda)})$ to $(\mathcal{M}_{n,k}, \omega_{n,k}^{(1)})$.

Proof. A direct substitution gives

$$F_\lambda^* \omega_{n,k}^{(1)} = \frac{i}{2} \sum_{j=1}^{n-1} dX_j \wedge d\overline{X}_j + \lambda \frac{i}{2} \sum_{\ell=0}^{k-1} dY_\ell \wedge d\overline{Y}_\ell = \omega_{n,k}^{(\lambda)}.$$

$\square$

This is an equivalence of unmarked ambient symplectic spaces. It rescales the musical embedding: for $\lambda \neq 1$, Parseval prevents $\sqrt{\lambda}Y_p$ from being a cardinality- k pattern spectrum. Thus λ remains a modeling parameter when the embedded seed or metric normalization is held fixed.

7.2. Action-angle coordinates. Let

$$U_{n,k} = \{(X, Y) \in \mathcal{M}_{n,k} : X_j \neq 0 \text{ for all } j, Y_\ell \neq 0 \text{ for all } \ell\}.$$

On $U_{n,k}$ write

$$X_j = \sqrt{2I_j} e^{i\theta_j}, \quad Y_\ell = \sqrt{2J_\ell} e^{i\phi_\ell},$$

with $I_j, J_\ell > 0$ and $\theta_j, \phi_\ell \in \mathbb{R}/2\pi\mathbb{Z}$.

Theorem 7.4 (Full layered symplectic form). *On $U_{n,k}$ one has*

$$\omega_{n,k}^{(\lambda)} = \sum_{j=1}^{n-1} dI_j \wedge d\theta_j + \lambda \sum_{\ell=0}^{k-1} dJ_\ell \wedge d\phi_\ell. \quad (7.2)$$

Proof. This is the standard polar-coordinate identity

$$\frac{i}{2} dz \wedge d\bar{z} = d\left(\frac{1}{2}|z|^2\right) \wedge d(\arg z)$$

applied to each coordinate separately and summed. $\square$

Equation (7.2) is the *full layered symplectic form*: the content actions I_j and angles θ_j occupy the first layer, while the order actions J_ℓ and angles ϕ_ℓ occupy the second. The actions (I_j, J_ℓ) extend smoothly across zero modes, but the angles (θ_j, ϕ_ℓ) are undefined at zero modes. Thus action-angle coordinates are used only on the nonvanishing-mode locus.

7.3. Completely integrable structure.

Theorem 7.5 (Complete integrability). *On each stratum $\mathcal{M}_{n,k}$, the functions*

$$I_1, \dots, I_{n-1}, J_0, \dots, J_{k-1}$$

pairwise Poisson commute and are functionally independent on the dense open set $U_{n,k}$. Thus the chosen action map (I, J) defines a completely integrable system on this ambient space; its positive regular fibers are tori. This does not prescribe musical dynamics on the finite seed.

Proof. In action-angle coordinates, the Poisson bracket is

$$\{F, G\} = \sum_{j=1}^{n-1} \left(\frac{\partial F}{\partial I_j} \frac{\partial G}{\partial \theta_j} - \frac{\partial F}{\partial \theta_j} \frac{\partial G}{\partial I_j} \right) + \frac{1}{\lambda} \sum_{\ell=0}^{k-1} \left(\frac{\partial F}{\partial J_\ell} \frac{\partial G}{\partial \phi_\ell} - \frac{\partial F}{\partial \phi_\ell} \frac{\partial G}{\partial J_\ell} \right).$$

Each I_j and J_ℓ depends only on itself, so all pairwise brackets vanish. Independence on $U_{n,k}$ is obvious from the coordinate system. $\square$

Corollary 7.6 (Autocorrelation observables Poisson commute). *The extended autocorrelation observables obtained from the inverse Fourier formulas for $A_S(\tau)$ and $C_p(r)$ are functions of the commuting actions $\{I_j\}$ and $\{J_\ell\}$ respectively. Therefore any two such observables Poisson commute on the thickened space.*

Proof. By theorems 3.2 and 5.3, both families are inverse Fourier transforms of the squared Fourier moduli, hence are functions only of the actions. $\square$

The ambient extensions are

$$A_I(r) = \frac{1}{n} \left(k^2 + 2 \sum_{j=1}^{n-1} I_j \eta_n^{jr} \right), \quad C_J(r) = \frac{2}{k} \sum_{\ell=0}^{k-1} J_\ell \zeta_k^{\ell r}.$$

For arbitrary ambient actions $I_{n-j} = I_j$ need not hold, so A_I can be complex and need not be an integer content count. These extensions and their real/imaginary parts Poisson commute. The action flows are complete coordinate rotations (speeds 1 and $1/\lambda$); any smooth Hamiltonian $h(I, J)$ also has complete flows since its actions and angular speeds stay constant on each trajectory.

8. HAMILTONIAN AND ANTI-SYMPLECTIC SYMMETRIES

We adopt the convention that a moment map μ for a Hamiltonian group action satisfies

$$\iota_{\xi_M} \omega = -d\langle \mu, \xi \rangle.$$

With the opposite sign convention all moment maps below change sign. The specified integer weights select particular circle extensions; agreement with the finite musical actions alone does not uniquely determine a continuous extension. Generic circle orbits leave the discrete seed.

8.1. Independent torus actions. The product torus

$$\mathbb{T}^{n-1} \times \mathbb{T}^k$$

acts on $\mathcal{M}_{n,k}$ by independent phase rotations in each coordinate:

$$(a, b) \cdot (X, Y) = ((a_j X_j)_{j=1}^{n-1}; (b_\ell Y_\ell)_{\ell=0}^{k-1}).$$

Its moment map is

$$\mu(X, Y) = ((I_j)_{j=1}^{n-1}, (\lambda J_\ell)_{\ell=0}^{k-1}).$$

Thus the nonnegative coordinates underlying the content autocorrelation, and the λ -scaled order-action coordinates underlying the order autocorrelation, are moment-map coordinates.

8.2. Transposition.

Definition 8.1 (Hamiltonian transposition action). Define an S^1 -action τ on $\mathcal{M}_{n,k}$ by

$$\tau_\alpha(X, Y) = ((e^{-j\alpha} X_j)_{j=1}^{n-1}; (e^{i\alpha} Y_\ell)_{\ell=0}^{k-1}), \quad (8.1)$$

where $\alpha \in \mathbb{R}/2\pi\mathbb{Z}$.

When $\alpha = 2\pi t/n$, this restricts to the discrete pitch transposition T_t on the embedded musical seed.

Theorem 8.2 (Transposition is Hamiltonian). *The action τ is Hamiltonian with moment map*

$$H_T = -\sum_{j=1}^{n-1} j I_j + \lambda \sum_{\ell=0}^{k-1} J_\ell. \quad (8.2)$$

Proof. The infinitesimal generator of (8.1) is

$$\mathcal{X}_T = -\sum_{j=1}^{n-1} j \frac{\partial}{\partial \theta_j} + \sum_{\ell=0}^{k-1} \frac{\partial}{\partial \phi_\ell}.$$

Contracting with (7.2) gives

$$\iota_{\mathcal{X}_T} \omega_{n,k}^{(\lambda)} = \sum_{j=1}^{n-1} j dI_j - \lambda \sum_{\ell=0}^{k-1} dJ_\ell = -dH_T.$$

This calculation holds on the dense nonvanishing-mode locus; both sides are smooth globally, so the identity extends across zero modes. $\square$

8.3. Cyclic reindexing.

Definition 8.3 (Hamiltonian cyclic reindexing action). Define an S^1 -action ρ on $\mathcal{M}_{n,k}$ by

$$\rho_\beta(X, Y) = (X; (e^{\ell i \beta} Y_\ell)_{\ell=0}^{k-1}).$$

When $\beta = 2\pi r/k$, this restricts to cyclic index shift by r on the order signal, as in [proposition 5.4](#).

Theorem 8.4 (Cyclic reindexing is Hamiltonian). *The action ρ is Hamiltonian with moment map*

$$H_R = \lambda \sum_{\ell=0}^{k-1} \ell J_\ell.$$

Proof. The infinitesimal generator is

$$\mathcal{X}_R = \sum_{\ell=0}^{k-1} \ell \frac{\partial}{\partial \phi_\ell}.$$

Thus

$$\iota_{\mathcal{X}_R} \omega_{n,k}^{(\lambda)} = -\lambda \sum_{\ell=0}^{k-1} \ell dJ_\ell = -dH_R.$$

The identity extends across zero modes by smoothness, as for transposition. $\square$

The commuting Hamiltonians H_T and H_R capture two different layers of symmetry. The restriction of τ to the n th roots of unity models pitch transposition exactly. The action ρ is a continuous Hamiltonian extension of cyclic reindexing; the actual root-forgetting operation in the discrete musical seed is recovered from the finite subgroup

$$C_k = \left\{ \rho_{2\pi r/k} : 0 \leq r \leq k-1 \right\} \subset S^1,$$

not from quotienting by the full circle. This distinction is essential in the orbifold analysis of [appendix A](#).

8.4. Inversion.

Definition 8.5 (Inversion involution). Define

$$\mathcal{I}(X_1, \dots, X_{n-1}; Y_0, \dots, Y_{k-1}) = (\overline{X_1}, \dots, \overline{X_{n-1}}; \overline{Y_0}, \overline{Y_{k-1}}, \dots, \overline{Y_1}).$$

Equivalently, in terms of order indices modulo k ,

$$(\mathcal{I}Y)_\ell = \overline{Y_{-\ell}}.$$

Theorem 8.6 (Inversion is anti-symplectic). *One has*

$$\mathcal{I}^2 = \text{id}, \quad \mathcal{I}^* \omega_{n,k}^{(\lambda)} = -\omega_{n,k}^{(\lambda)}.$$

Proof. Complex conjugation on each complex coordinate sends $\frac{i}{2} dz \wedge d\bar{z}$ to its negative. Permutation of coordinates does not affect the form. Therefore each summand in [\(7.1\)](#) changes sign, and the full form does as well. $\square$

Remark 8.7. This is the mathematically correct reason that the full T/I quotient is not globally symplectic in the naive sense: transposition is symplectic and Hamiltonian, but inversion reverses the symplectic form. The natural object is therefore a symplectic quotient by the transposition subgroup together with an anti-symplectic involution descended from inversion.

9. REDUCTION, FIBERS, AND STRATIFICATION

9.1. Symplectic reduction by transposition. Work on $V = \{(X, Y) : Y \neq 0\}$, where the weight-one action on every order coordinate makes transposition free. Every nonempty level of $H_T|_V$ is regular. Marsden–Weinstein reduction therefore gives a smooth symplectic manifold $(H_T^{-1}(c) \cap V)/S^1$ of dimension $2n + 2k - 4$.

All cardinality- k musical seeds lie on the single level

$$c_{n,k,\lambda} = \frac{\lambda k^2}{2} - \frac{nk(n-k)}{4}.$$

Indeed Parseval gives $\sum_\ell |Y_\ell|^2 = k^2$ and $\sum_{j \neq 0} |X_j|^2 = nk - k^2$; content reality gives $I_{n-j} = I_j$, so $\sum_j j I_j = nk(n-k)/4$. Other levels contain no such seeds. Equality of two seed S^1 orbits forces a discrete n th-root phase by inverse order DFT, and hence a discrete transposition. Thus discrete transposition classes embed injectively in this distinguished reduction, including when $c_{n,k,\lambda} = 0$.

Forgetting the cyclic root is the finite C_k quotient, not reduction by the whole reindexing circle. Unreduced seed reindexing is free, but after transposition reduction its stabilizer is the group of periods of the directed gap word. A constant-gap cycle can have full residual C_k isotropy. Continuous reindexing reduction is a different operation.

9.2. Vector-bundle interpretation. The projection

$$\pi_{n,k} : \mathcal{M}_{n,k} \rightarrow \mathbb{C}^{n-1}, \quad \pi_{n,k}(X, Y) = X$$

is a trivial complex vector bundle with fiber $\mathbb{C}^k$. The symplectic form splits as a product form:

$$\omega_{n,k}^{(\lambda)} = \omega_{\text{content}} \oplus \lambda \omega_{\text{order}}.$$

Accordingly, each cardinality- k stratum is a *content base* together with an attached *order fiber*. This is the precise mathematical version of the intuitive statement “attach each local permutation fiber to its content-coordinate point.”

Proposition 9.1 (Local permutation fibers). *Fix $X \in \mathbb{C}^{n-1}$. The fiber $\pi_{n,k}^{-1}(X)$ is symplectomorphic to $(\mathbb{C}^k, \lambda \omega_{\text{std}})$. The intersection*

$$\pi_{n,k}^{-1}(X) \cap \iota_{n,k}(\mathcal{P}_{n,k})$$

is a finite set consisting of the discrete rooted orderings compatible with the content coordinates X . If X is the spectrum of a cardinality- k content S ,

this intersection has $k!$ elements; fixing $r \in S$ or quotienting by C_k gives $(k-1)!$. Otherwise the intersection is empty.

Proof. The first claim is immediate from the product decomposition. For the second, $\iota_{n,k}(\mathcal{P}_{n,k})$ is discrete by construction, and fixing X fixes the content layer. If X is not in the image of any content set, the intersection is empty. If $X = X_S$ for a content set S , then the intersection with $\iota_{n,k}(\mathcal{P}_{n,k})$ consists of the $k!$ rooted ordered tuples with content S , before any root-fixing or cyclic quotient. $\square$

9.3. Stratification across cardinalities. Different cardinalities have different order-fiber dimensions; we equip their union with the disjoint-union topology.

Definition 9.2 (Total layered space). Define

$$\mathcal{M}_n = \bigsqcup_{k=1}^n \mathcal{M}_{n,k}.$$

Proposition 9.3 (Cardinality grading). *With the disjoint-union topology, $\mathcal{M}_n$ is a cardinality-graded family of symplectic manifolds of dimensions $2(n-1+k)$. For $n \geq 2$ it is not a fixed-dimensional manifold. Its cardinality decomposition is a trivial stratification with empty cross-component frontier relations.*

Proof. Each component is open and closed and carries its specified symplectic form. The dimensions vary with k ; no component meets another's closure. Conventions permitting different dimensions on different connected components may call the union a manifold. No attachment topology is supplied here. $\square$

The one-note add/remove graph is a separate combinatorial construction; it is not an attachment relation of this disjoint union.

10. DISCUSSION: REPRESENTATIONS OF DISTANCE AND ORDER

The historical “metric ladder” names four useful representations, but they do not form a chain of increasing information. A labeled distance matrix determines its all-pairs distance multiset and, given a traversal, that traversal's unsigned step sequence. Forgetting a step sequence's reading order gives its step multiset, not its all-pairs distance multiset. For example, $(0, 1, 3, 6)$ and $(0, 1, 4, 6)$ have the same directed gap multiset $[1, 2, 3, 6]$ in G_{12} , but their unordered pairwise circular-distance multisets are respectively $[1, 2, 3, 3, 5, 6]$ and $[1, 2, 3, 4, 5, 6]$.

Directed cyclic gaps are differences in G_n , not unsigned metric distances. With a root and the closure/distinctness conditions of [proposition 2.7](#), they reconstruct the pattern. Without a root they determine it up to transposition, with order indices still retained. Discarding traversal order also

identifies different orderings of the same content, not merely homometric contents.

The directed autocorrelation $A_S(r)$ counts ordered pairs with difference r , including $A_S(0) = |S|$. In twelve tones let $IV_S(d)$ count unordered distinct pairs at circular distance d and $F_S(d)$ count ordered distinct pairs at that distance. Then

$$\begin{aligned} F_S(d) &= 2IV_S(d), & A_S(6) &= 2IV_S(6), \\ A_S(d) &= A_S(12-d) = IV_S(d) & (1 \leq d < 6). \end{aligned}$$

Thus cardinality together with IV_S determines A_S , but the six-bin folded histogram is not the twelve-bin signal in [theorem 3.2](#). Homometry means equality of directed autocorrelations, including trivial equalities and transpositions/inversions. Z-relatedness additionally excludes T/I equivalence. The sets $\{0, 1, 4, 6\}$ and $\{0, 1, 3, 7\}$ are a nontrivial example, both with $IV = (1, 1, 1, 1, 1, 1)$.

The automorphism group is derived from a metric; it does not in general recover it (scaling all nonzero distances preserves every automorphism). Ambient T/I symmetries induce a subgroup of metric automorphisms of a content set. The explorer's `symmetry_order` counts transpositions preserving the chosen set, not all T/I symmetries or all metric automorphisms.

RMCP supplies redundant encodings and implementation checks of candidate mathematical objects. Its legacy `cyclic_autocorrelation` field stores F , and its `homometric` function tests nontrivial Z-relatedness. These names do not change the preceding conventions. Prefix/suffix compression and shuffled baselines are operational statistics, not theorems of self-teaching, semantic recovery, or intentional communication. The published 14,262 orderings belong to 216 sampled/enumerated fibers; they are not all orderings of the full 223-class universe. Detector hits in repertoire require controlled interpretation and do not establish intentionality. No general probability or perceptual utility guarantee follows from the Fourier embedding.

11. VISUALIZATION THEOREMS

The exact object lives in high-dimensional symplectic coordinates, but one may obtain controlled finite-dimensional renderings by projecting the content base and shrinking the order fibers.

11.1. Four-dimensional direct-sum renderings. Choose a map $\Phi : \mathcal{B} \rightarrow \mathbb{R}^2$ from the finite set $\mathcal{B}$ of content-coordinate points, or from a specified fitted base model; examples include multidimensional scaling or diffusion geometry applied to a chosen content metric [14, 4]. Choose a linear or nonlinear map

$$\Psi : \mathbb{C}^k \rightarrow \mathbb{R}^2$$

representing the order fiber. For $\varepsilon > 0$ define

$$\mathcal{V}_\varepsilon^{(4)}(X, Y) = (\Phi(X), \varepsilon\Psi(Y)) \in \mathbb{R}^4.$$

This is a chosen direct-sum rendering: two coordinates for content and two for the attached order fiber. Rotating $\mathbb{R}^4$ and projecting to $\mathbb{R}^3$ or $\mathbb{R}^2$ produces the animated “4D fiber bundle” pictures used in practice.

11.2. Three-dimensional frame renderings. For each point b_i of a finite rendered base choose an isometric linear map $E_i : \mathbb{R}^2 \rightarrow \mathbb{R}^3$. Place a fiber point at $b_i + \varepsilon E_i \Psi_i(Y)$. These chosen planes are rendering data, not tangent planes of the full high-dimensional content space. For direct-sum rendering, injectivity on the finite sample is guaranteed by injective base and fiber maps and $\varepsilon \neq 0$; further projection can introduce collisions.

Proposition 11.1 (Small-radius separation). *Let $B = \{b_1, \dots, b_N\} \subset \mathbb{R}^m$ have $N \geq 2$ distinct base points and let $F_i \subset \mathbb{R}^m$ be nonempty bounded fiber images with*

$$R = \max_i \sup_{u \in F_i} \|u\| < \infty, \quad \delta = \min_{i \neq j} \|b_i - b_j\| > 0.$$

If $\varepsilon \geq 0$ and $2\varepsilon R < \delta$, then the translated sets

$$b_i + \varepsilon F_i$$

are pairwise disjoint.

Proof. If $R = 0$, the conclusion is immediate. Otherwise assume $R > 0$. If $u \in F_i$ and $v \in F_j$, then

$$\|(b_i + \varepsilon u) - (b_j + \varepsilon v)\| \geq \|b_i - b_j\| - \varepsilon \|u - v\| \geq \delta - 2\varepsilon R > 0.$$

Hence no two translated fiber images intersect. $\square$

This estimate separates the images of different fibers when the rendered bases are distinct. It gives a sufficient geometric separation condition; visual resolution and collisions within a fiber require separate checks.

For zero or one base point separation is vacuous. Shrinking fibers only separates distinct bases; it cannot repair collisions within a fiber. For a finite distance matrix D , exact Euclidean realization in $\mathbb{R}^2$ is possible if and only if $-\frac{1}{2}H(D^{\circ 2})H$ is positive semidefinite of rank at most two, where $H = I - N^{-1}\mathbf{1}\mathbf{1}^T$ is the centering matrix for N points. Otherwise the plot is an approximation; no rank or distortion certificate for the supplied plots is asserted here.

12. WORKED CONSEQUENCES IN THE TWELVE-TONE CASE

For $n = 12$, the above formalism includes, without conceptual change:

- all scale and chord contents as subsets of $\mathbb{Z}_{12}$, carrying Forte labels (e.g., 3-11 for the major/minor triad, 7-35 for the diatonic scale);
- all ordered nonrepeating patterns, including segments of twelve-tone rows;
- harmonic extensions as larger content sets containing a given subset (e.g., the dominant seventh 4-27 extends the major triad 3-11);

- Slonimsky interval cycles as periodic gap signals (e.g., equal-step cycles like the diminished seventh 4-28 and whole-tone collection 6-35);
- and local permutation fibers of size $(k-1)!$ for a chosen labeled content representative and a fixed root in it, equivalently cyclic orders before any content-stabilizer quotient.

Figures 1 to 5 illustrate several quotient-level visualizations: the first three emphasize structure-aware shell and rosette maps, while the last two reproduce earlier approximate metric renderings. These illustrate selected representations associated with the construction. Their metric fidelity and sampled-fiber topology are not consequences of the abstract embedding or separation estimate.

Structure-aware renderings. Figure 1 shows the 223 nonempty T/I -classes of $\mathbb{Z}_{12}$ as a stratified shell graph: nodes are arranged by cardinality (radius), colored by cardinality, and sized by the order of the transposition stabilizer; edges connect classes that differ by the addition or removal of a single pitch class. This layout makes the cardinality grading of [proposition 9.3](#) visually explicit and reveals the dense internal connectivity of the base quotient graph. Landmark classes are labeled by their common names (“Dim7” = Forte 4-28, “Whole-tone” = 6-35, “Octatonic” = 8-28, “Aug triad” = 3-12, “Aggregate” = 12-1).

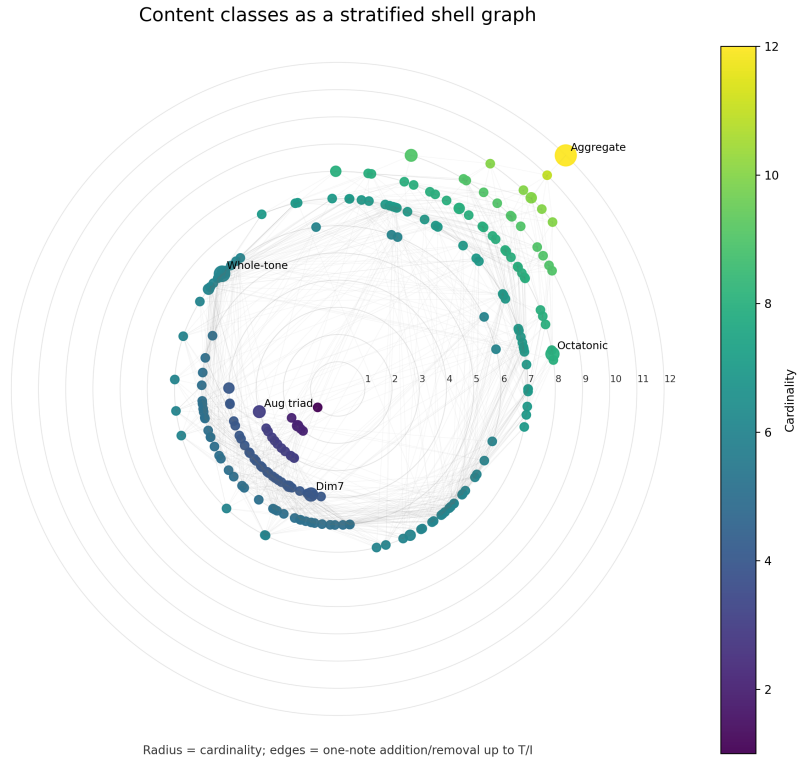


FIGURE 1. Content classes as a stratified shell graph for $n = 12$. Radius encodes cardinality; edges represent one-note addition/removal up to T/I -equivalence. Node size reflects the number of content-preserving transpositions, making landmarks such as the whole-tone collection and diminished seventh chord visually prominent.

Attaching local permutation fibers to this shell graph yields a *rosette bundle map* in which each content node sprouts a small rosette of fiber points. Figure 2 shows this construction with fiber vertices colored by the cyclic step-word entropy $H(p)$ described in remark 4.4. The entropy coloring exposes internal fiber structure at a glance: blue vertices have lower step-histogram diversity, while yellow-green vertices have higher diversity.

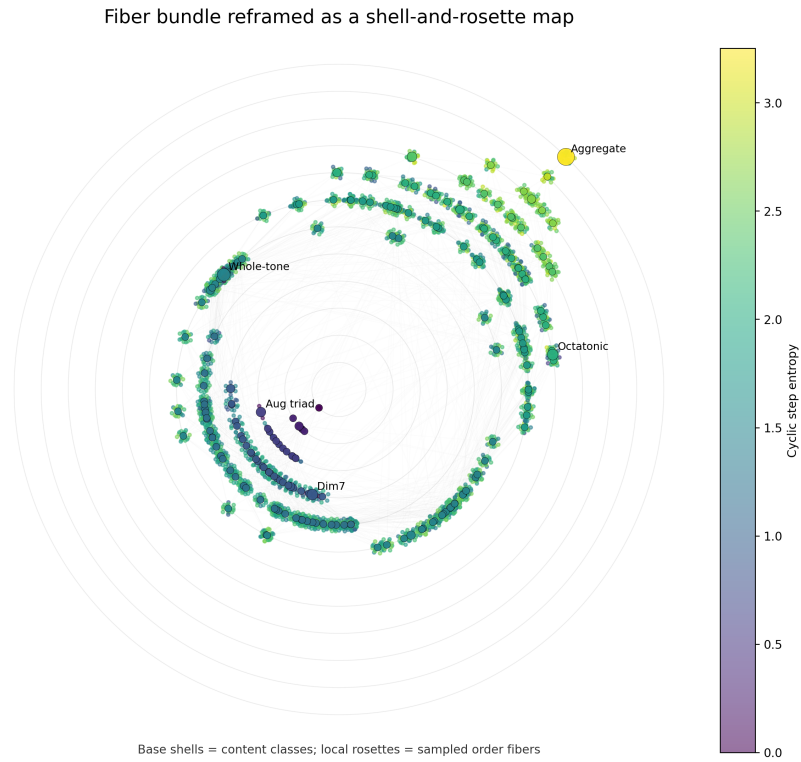


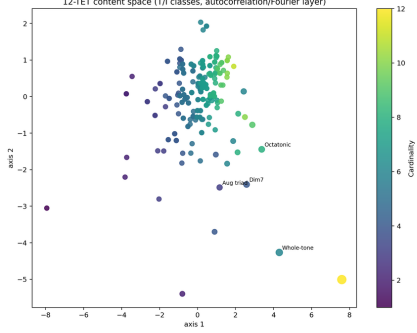
FIGURE 2. The fiber bundle reframed as a shell-and-rosette map. Base shells encode content classes; local rosettes represent sampled order fibers. Color encodes cyclic step entropy: blue indicates lower step diversity, yellow higher step diversity. This rendering is the structure-aware alternative to the generic 4D scatter figures.

Figure 3 places these structure-aware renderings alongside earlier approximate metric renderings.

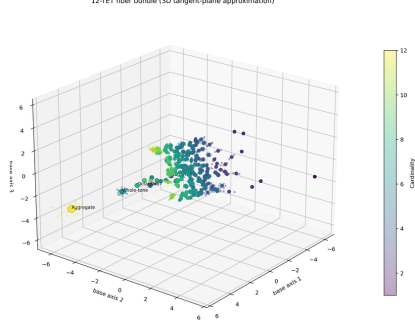
From point cloud to structured map

The shell graph exposes stratification; the rosette map exposes local fibers.

Current content MDS

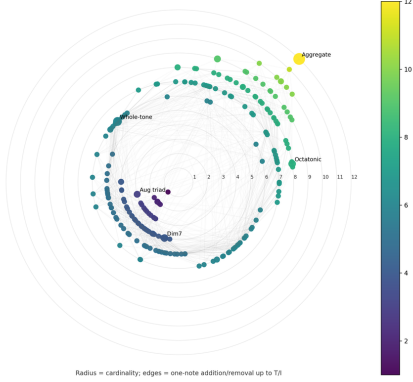


Current bundle scatter



Shell graph (recommended base view)

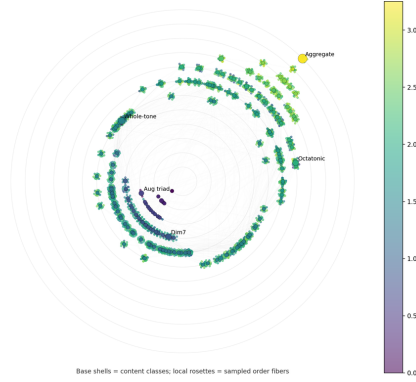
Content classes as a stratified shell graph



Radius = cardinality; edges = one-note addition/removal up to T/I

Rosette bundle (recommended public view)

Fiber bundle reframed as a shell-and-rosette map



Base shells = content classes; local rosettes = sampled order fibers

FIGURE 3. From point cloud to structured map. Top row: MDS content scatter and 3D bundle scatter from the earlier prototype. Bottom row: stratified shell graph (left) and rosette bundle map with entropy coloring (right). The shell graph exposes the cardinality stratification; the rosette map exposes local fiber structure.

Supplementary metric renderings. For completeness, figures 4 and 5 reproduce the MDS-based content scatter and 4D direct-sum projection from the earlier computational prototype. These approximate metric relationships without a certified distance-preservation guarantee and do not expose the combinatorial graph skeleton; they are therefore best viewed as supplements to the shell graph and rosette map above.

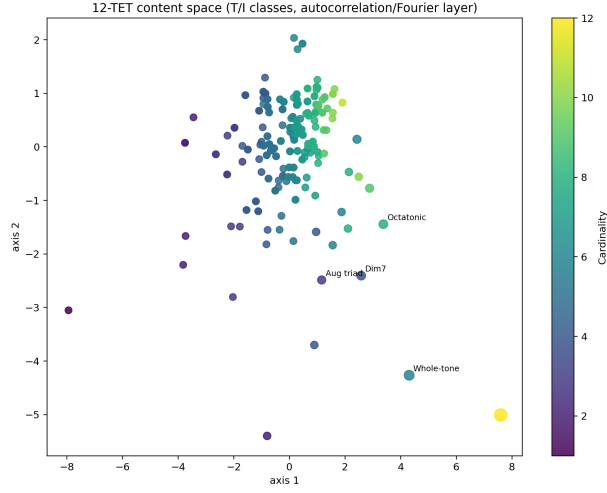


FIGURE 4. Two-dimensional content rendering via multidimensional scaling. This diagram approximates content distances but does not expose the cardinality stratification visible in figure 1.

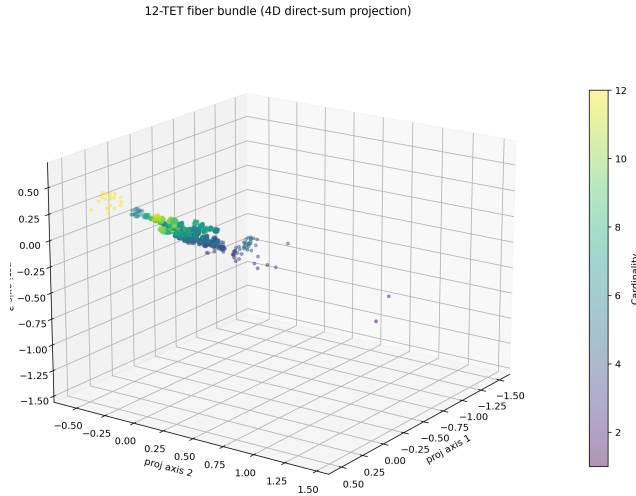


FIGURE 5. Projected 4D direct-sum rendering of the content base with attached order fibers. The picture uses chosen content representatives and sampled rooted traversals. The ambient family $\mathcal{M}_{12} = \bigsqcup_{k=1}^{12} \mathcal{M}_{12,k}$ motivates the separation into layers; this projection does not certify preservation of its geometric structure.

Computational anatomy of the fiber space. The fiber construction is not merely theoretical. A companion computational pipeline generates exact or sampled fibers for every content class with three or more pitch classes (cardinalities 1 and 2 have trivial single-element fibers). [Table 1](#) summarizes the combinatorial landscape.

TABLE 1. Fiber coverage by cardinality for the $n = 12$ case. The table counts rooted cyclic orderings per chosen representative before quotienting by content stabilizers. For every content class of cardinality k , the exact count of rooted cyclic orderings is $(k - 1)!$ (by [proposition 2.5](#)), independent of the symmetry order of the T/I -class. At cardinality 6 and above, the visualization samples at most 96 orderings per fiber.

Cardinality	T/I -classes	Orderings/fiber	Sampling	Total orderings
1	1	1	—	—
2	6	1	—	—
3	12	2	exhaustive	24
4	29	6	exhaustive	174
5	38	24	exhaustive	912
6	50	≤ 96	sampled from 120	4,800
7	38	≤ 96	sampled from 720	3,648
8	29	≤ 96	sampled from 5,040	2,784
9	12	≤ 96	sampled from 40,320	1,152
10	6	≤ 96	sampled from 362,880	576
11	1	≤ 96	sampled from 3,628,800	96
12	1	≤ 96	sampled from 39,916,800	96

The combinatorial explosion is dramatic: from 2 orderings at cardinality 3 to tens of millions at cardinality 12. Note that the rooted cyclic fiber count $(k - 1)!$ does not depend on the symmetry order of the T/I -class: fixing a root selects a unique cyclic reading. The diminished seventh chord (4-28, transposition-stabilizer order 4) has the same $6 = (4 - 1)!$ rooted orderings as any other tetrachord. Symmetry enters a further quotient through the action of the content stabilizer on cyclic orders. The number of distinct transposed contents is a different count. Similarly, the whole-tone collection (6-35, transposition-stabilizer order 6) has the full $(6 - 1)! = 120$ rooted orderings. Further quotienting by content stabilizers requires a separate orbit count; the stabilizer action on cyclic orders can itself have fixed points, so one cannot simply divide $(k - 1)!$ by the stabilizer order. The sampled graph retains adjacent-swap edges between selected vertices; no preservation theorem for full fiber topology is asserted.

The full dataset contains 223 content classes, 1,051 add/remove edges, and 216 computed fibers (the seven classes of cardinality $k \leq 2$ have trivial single-element fibers and are excluded from nontrivial fiber computation)

with a total of 14,262 orderings and their associated swap-edge graphs. Each content class is identified both by its Forte number and, where one exists, its common Western name (see [remark 2.2](#)).

Worked musical case studies. We present three worked examples illustrating different corners of the framework.

Tonal: the major triad and its extensions. Let $S = \{0, 4, 7\} \subset G_{12}$ (Forte 3-11, the major/minor triad). The content spectrum is

$$X_j = 1 + \eta_{12}^{-4j} + \eta_{12}^{-7j}, \quad 1 \leq j \leq 11.$$

The largest squared magnitudes are $|X_3|^2 = |X_9|^2 = 5$, followed by $|X_5|^2 = |X_7|^2 = 2 + \sqrt{3}$ and $|X_4|^2 = |X_8|^2 = 3$. Adding a note q changes every coefficient by η_{12}^{-jq} and can change this ranking: for the dominant seventh $\{0, 4, 7, 10\}$, the largest values are instead $|X_4|^2 = |X_8|^2 = 7$. Spectral proximity between these contents therefore requires a specified metric and normalization; it does not follow from their inclusion relation.

The rooted cyclic fiber has $(3-1)! = 2$ orderings: $(0, 4, 7)$ and $(0, 7, 4)$, with gap words $(4, 3, 5)$ and $(7, 9, 8)$ respectively. The first is the familiar ascending arpeggiation; the second is the descending arpeggiation read as a cyclic pattern. The step-word entropy of $(4, 3, 5)$ is $H = \log_2 3 \approx 1.585$ bits (maximal step diversity for three gaps); $(7, 9, 8)$ has the same entropy $H = \log_2 3 \approx 1.585$ bits, since each gap word has three distinct step classes occurring once. Both orderings sit in the same small fiber, connected by a single adjacent swap, illustrating that at cardinality 3 the adjacent-swap graph on the two non-root entries is a single edge.

Symmetry-rich: the diminished seventh chord. Let $S = \{0, 3, 6, 9\}$ (Forte 4-28, the diminished seventh chord). The stabilizer $\text{Stab}_T(S) = \{0, 3, 6, 9\} \cong C_4$, so this set is invariant under transposition by any minor third. The content spectrum is supported, apart from the zeroth mode, on $j = 4, 8$: the coefficients $|X_4| = |X_8| = 4$ dominate, reflecting the C_4 -symmetry of the diminished seventh (invariance under transposition by 3). All other magnitudes $|X_j|$ vanish for $j \not\equiv 0 \pmod{4}$. Because $\text{Stab}_T(S)$ has order 4, there are only $12/4 = 3$ distinct transpositions of the diminished seventh, compared with 12 for a generic tetrachord.

The rooted cyclic fiber has $(4-1)! = 6$ orderings. The equal-step ordering $(0, 3, 6, 9)$ has gap word $(3, 3, 3, 3)$ with entropy $H = 0$ (perfectly even). The ordering $(0, 6, 3, 9)$ has gap word $(6, 9, 6, 3)$ with entropy $H \approx 1.5$ bits. Thus this fiber contains different step-histogram diversities; entropy alone does not classify traversal smoothness. This example illustrates [corollary 4.2](#): the gap word $(3, 3, 3, 3)$ is periodic with period 1, so its gap spectrum is concentrated on a single mode.

Ordered/serial: a Slonimsky-type interpolated pattern. Consider the ordered pattern $p = (0, 1, 4, 5, 8, 9)$ on the content set $S = \{0, 1, 4, 5, 8, 9\}$ (Forte 6-20). The gap word is $(1, 3, 1, 3, 1, 3)$, which is periodic with period 2: it alternates a semitone step with a minor-third leap. By [theorem 4.1](#), the

gap spectrum $G_p(\ell)$ is supported on $\{0, 3\}$ (the multiples of $k/d = 6/2 = 3$), confirming sparse spectral support. This is exactly a Slonimsky interpolation: the skeleton is a major-third cycle $(0, 4, 8)$, and each skeleton note is followed by its chromatic upper neighbor.

For $S = \{0, 1, 4, 5, 8, 9\}$,

$$X_j = (1 + \eta_{12}^{-j})(1 + \eta_{12}^{-4j} + \eta_{12}^{-8j}).$$

The second factor vanishes unless $j \equiv 0 \pmod{3}$, and the first factor vanishes at $j = 6$. Hence, among $1, \dots, 11$, the only nonzero content modes are $j = 3, 9$. The gap spectrum is supported on $\ell = 0, 3$, so the auxiliary gap layer detects the alternating interval cycle. The content spectrum and the auxiliary gap spectrum both reflect the threefold major-third backbone, while the faithful order spectrum records the rooted ordering itself. This spectral coherence between content and gap diagnostics is exactly the kind of structural alignment that the layered framework is designed to detect.

Interactive companion. An interactive realization of these renderings—the *Layered Bundle Explorer*—accompanies this paper as a web-based computational artifact. It implements all four views discussed above (MDS scatter, 4D projection, shell graph, rosette map), together with semantic zoom for progressive detail revelation, a configurable hop-radius edge filter, cardinality band selection, and on-demand fiber inspection with entropy coloring and hover tooltips. Selecting any node displays its Forte number and common name alongside its pitch-class content, fiber statistics, and step-word data. A guided-tour mode animates a traversal through musically meaningful paths—from triads through diminished ladders to whole-tone and octatonic regions—offering a narrated introduction to the sampled graph and fiber views. In addition, a grammar-learnability overlay scores every one of the 14,262 fiber orderings using the compression of later cyclic laps by a grammar fitted to earlier laps, coloring fibers by that measured prefix/suffix compression statistic alongside the entropy channel (see the discussion in [section 10](#)). The explorer is available at the companion project website.

13. CONCLUSION

The construction separates three tasks that require different kinds of justification: representing finite musical data without loss, extracting invariants that deliberately identify some objects, and choosing a continuous ambient geometry. Directed gaps with a root and full order Fourier coordinates solve reconstruction problems. Autocorrelations summarize interval relationships. The layered embedding retains both content and order coordinates so that these observables can be studied together, even though the order factor alone already determines the pattern.

The weighted standard form makes the chosen ambient model symplectic. Its phase rotations extend the finite musical symmetries, but generic continuous trajectories leave the musical seed. Transposition reduction and finite

cyclic root-forgetting are distinct operations; residual isotropy depends on the traversal as well as its content. Inversion reverses the form, so its further identification must be treated separately from a symplectic quotient. Across cardinalities the topology is the specified disjoint union; the add/remove graph supplies a separate relation among content classes.

The contribution is a compatible assembly of these constructions with explicit conventions, proofs, and a machine-checked discrete foundation. The standard Fourier and symplectic ingredients are not claimed as new. Low-dimensional plots make selected aspects of the model accessible: shell graphs display adjacency, local fiber plots display sampled traversals, and metric embeddings approximate chosen distances. Their separation and fidelity depend on the stated hypotheses and on numerical validation, rather than following automatically from the existence of the ambient form.

Further work should examine which observables assist analysis or composition, measure rendering distortion, and test empirical claims against suitable controls. Register, repeated notes, timing, and voice-leading constraints would require extensions of the present domain; comparison with the chord spaces of Tymoczko and Callender–Quinn–Tymoczko is therefore a research direction rather than an identification of the models [16, 3]. Further formalization can address the paper’s periodicity and higher-geometric results. The present model supplies a precise foundation for those investigations without making their outcomes assumptions of the theory.

APPENDIX A. SINGULAR STABILIZERS AND REDUCED-ORBIFOLD STRATA

This appendix isolates the finite stabilizers that control the singular loci of the quotient constructions. Two kinds of isotropy occur naturally: discrete stabilizers in the content quotients and finite stabilizers of the cyclic reindexing action on the thickened order fiber.

A.1. Content stabilizers in the discrete quotient.

Definition A.1. For a nonempty set $S \subset G_n$ define its transposition and T/I stabilizers by

$$\text{Stab}_T(S) = \{t \in G_n : S + t = S\}, \quad \text{Stab}_{TI}(S) = \{g \in D_n : g(S) = S\}.$$

For $t \in G_n$ we write

$$I_t(x) = t - x$$

for the affine inversion centered at $t/2$.

Proposition A.2. *The set $\text{Stab}_T(S)$ is a subgroup of G_n . Moreover S is a union of cosets of $\text{Stab}_T(S)$, and conversely any subgroup $H \leq G_n$ with respect to which S is a union of cosets satisfies $H \leq \text{Stab}_T(S)$. In particular, $\text{Stab}_T(S)$ is nontrivial if and only if S is periodic under a nonzero translation.*

Proof. If $a, b \in \text{Stab}_T(S)$, then

$$S + (a - b) = (S + a) - b = S - b = S,$$

so $\text{Stab}_T(S)$ is a subgroup. For $s \in S$ and $h \in \text{Stab}_T(S)$ one has $s + h \in S$, hence every $\text{Stab}_T(S)$ -coset meeting S is contained in S ; therefore S is a union of such cosets. Conversely, if S is a union of cosets of a subgroup H , then $S + h = S$ for every $h \in H$, so $H \leq \text{Stab}_T(S)$. The final statement is immediate. $\square$

Proposition A.3. *Let $S \subset G_n$ be nonempty.*

- (i) *The affine inversion I_t stabilizes S if and only if $S = t - S$.*
- (ii) *If no affine inversion stabilizes S , then $\text{Stab}_{TI}(S) = \text{Stab}_T(S)$.*
- (iii) *If some affine inversion stabilizes S , then*

$$\text{Stab}_{TI}(S) = \text{Stab}_T(S) \rtimes \langle I_t \rangle$$

for any such t , and in particular $|\text{Stab}_{TI}(S)| = 2 |\text{Stab}_T(S)|$.

Proof. The first statement is simply the identity $I_t(S) = t - S$. For the second and third, note that every element of D_n has the form T_a or $T_a \circ I_0 = I_a$. Thus any stabilizer element with negative sign is an affine inversion preserving S . If no such inversion exists, only translations remain. Suppose now that $I_t(S) = S$. For every $h \in \text{Stab}_T(S)$ one has

$$I_t \circ T_h \circ I_t = T_{-h},$$

so $\text{Stab}_T(S)$ is normal in $\text{Stab}_{TI}(S)$. Every sign-reversing stabilizer is of the form $T_h \circ I_t$ for some $h \in \text{Stab}_T(S)$, because the product of two sign-reversing elements is a translation in $\text{Stab}_T(S)$. Hence $\text{Stab}_{TI}(S)$ is the claimed semidirect product and has twice the size of $\text{Stab}_T(S)$. $\square$

Remark A.4. These stabilizers describe the group actions presenting the discrete content quotients. The coarse quotient sets themselves are finite and discrete; their nontrivial stabilizers do not by themselves define geometric singular loci. In the reduced ambient quotient below, seed isotropy depends on the gap period of the traversal, not solely on its content stabilizer. For example, the whole-tone content admits both the constant-gap traversal $(0, 2, 4, 6, 8, 10)$ and $(0, 2, 6, 4, 8, 10)$, whose gaps $(2, 4, 10, 4, 2, 2)$ have full period. Their residual cyclic stabilizers are respectively C_6 and the trivial group.

A.2. Finite stabilizers in the order fiber. Let $C_k = \langle \rho \rangle$ be the cyclic group generated by the discrete reindexing element

$$\rho(Y_0, \dots, Y_{k-1}) = (Y_0, \zeta_k Y_1, \dots, \zeta_k^{k-1} Y_{k-1}), \quad \zeta_k = e^{2\pi i/k}.$$

This is the finite subgroup of the Hamiltonian circle action from [theorem 8.4](#) corresponding to actual cyclic reindexing of rooted patterns.

Proposition A.5. For $Y = (Y_0, \dots, Y_{k-1}) \in \mathbb{C}^k$, let

$$L(Y) = \{\ell \in \{0, \dots, k-1\} : Y_\ell \neq 0\}, \quad d(Y) = \gcd(\{k\} \cup L(Y)).$$

Then the stabilizer of Y under C_k is

$$\text{Stab}_{C_k}(Y) = \langle \rho^{k/d(Y)} \rangle \cong C_{d(Y)}.$$

In particular, Y has nontrivial stabilizer if and only if every nonzero Fourier mode of Y lies in the proper subgroup $d(Y)\mathbb{Z}/k\mathbb{Z}$.

Proof. The condition $\rho^r Y = Y$ is equivalent to

$$e^{2\pi i r \ell / k} Y_\ell = Y_\ell$$

for every ℓ , hence to

$$k \mid r\ell \quad \text{for all } \ell \in L(Y).$$

Write $d = d(Y)$ and $k = dk'$. Because every $\ell \in L(Y)$ is divisible by d , every multiple of k' belongs to the stabilizer: if $r = qk'$, then

$$r\ell = qk'\ell = qk \frac{\ell}{d}$$

is divisible by k . Conversely, write each $\ell \in L(Y)$ as $\ell = dm_\ell$. By definition of d , the integers k' and $\{m_\ell : \ell \in L(Y)\}$ are coprime as a family. Hence there exist integers a_ℓ and b such that

$$\sum_{\ell \in L(Y)} a_\ell m_\ell + bk' = 1.$$

If $k \mid r\ell$ for all $\ell \in L(Y)$, then $k' \mid rm_\ell$ for all such ℓ , so multiplying the Bézout identity by r shows $k' \mid r$. Therefore the stabilizer consists exactly of the classes r divisible by $k' = k/d$, which form a subgroup of order d . $\square$

Corollary A.6. Let $p \in \mathcal{P}_{n,k}$ and let Y be its order-spectrum coordinate in the embedding $\iota_{n,k}(p) = (X, Y)$. Then $\text{Stab}_{C_k}(Y)$ is trivial.

Proof. Assume $\text{Stab}_{C_k}(Y)$ is nontrivial. By [proposition A.5](#), there exists $d > 1$ dividing k such that $Y_\ell = 0$ whenever $d \nmid \ell$. Writing the inverse discrete Fourier transform,

$$s_p(m) = \frac{1}{k} \sum_{\ell=0}^{k-1} Y_\ell \zeta_k^{\ell m},$$

one obtains

$$s_p\left(m + \frac{k}{d}\right) = \frac{1}{k} \sum_{d \mid \ell} Y_\ell \zeta_k^{\ell m} \zeta_k^{\ell k/d} = \frac{1}{k} \sum_{d \mid \ell} Y_\ell \zeta_k^{\ell m} = s_p(m),$$

because $\zeta_k^{\ell k/d} = 1$ when $d \mid \ell$. Thus the order signal is periodic with period $k/d < k$. Since $s_p(m) = \eta_n^{p_m}$ and the map $x \mapsto \eta_n^x$ is injective on G_n , it follows that $p_{m+k/d} = p_m$ for all m , contradicting the requirement that the entries of p be distinct. $\square$

Remark A.7. The preceding corollary shows that the embedded rooted-pattern seed lies in the regular part of the finite cyclic quotient. This freeness concerns the unreduced action only; reduction by transposition can create residual seed isotropy. By contrast, the gap signal may possess genuine periodicity on the discrete seed, and that periodicity is precisely the Slonimsky phenomenon analyzed in [theorem 4.1](#).

A.3. Reduced orbifold strata. Let

$$V_{n,k} = \{(X, Y) \in \mathcal{M}_{n,k} : Y \neq 0\}$$

be the open locus on which the transposition action is free.

Lemma A.8. *The Hamiltonian transposition action τ from [theorem 8.2](#) is free on $V_{n,k}$.*

Proof. Suppose $\tau_\alpha(X, Y) = (X, Y)$ with $(X, Y) \in V_{n,k}$, i.e. $Y \neq 0$. Since every order coordinate has weight 1 in [equation \(8.1\)](#), there exists ℓ with $Y_\ell \neq 0$, and therefore

$$e^{i\alpha} Y_\ell = Y_\ell.$$

Hence $e^{i\alpha} = 1$, so $\alpha = 0$ in S^1 . □

For a regular value c of $H_T|_{V_{n,k}}$, define

$$Z_{n,k,c} = H_T^{-1}(c) \cap V_{n,k}, \quad \mathcal{R}_{n,k,c} = Z_{n,k,c}/S^1.$$

Theorem A.9. *Let c be a regular value of $H_T|_{V_{n,k}}$.*

- (i) $\mathcal{R}_{n,k,c}$ is a symplectic manifold.
- (ii) The finite group C_k acts symplectically on $\mathcal{R}_{n,k,c}$, and the quotient

$$\mathcal{Q}_{n,k,c} = \mathcal{R}_{n,k,c}/C_k$$

is a symplectic orbifold.

- (iii) For every subgroup $H \leq C_k$, the fixed-point set $\mathcal{R}_{n,k,c}^H$ is a symplectic submanifold. The orbit-type subset

$$\mathcal{R}_{n,k,c}(H) = \mathcal{R}_{n,k,c}^H \setminus \bigcup_{H \subsetneq K \leq C_k} \mathcal{R}_{n,k,c}^K$$

is open in $\mathcal{R}_{n,k,c}^H$, and its image in $\mathcal{Q}_{n,k,c}$ is a smooth symplectic orbit-type stratum; the ambient quotient has a local tangent chart modulo H .

Proof. By [lemma A.8](#), the S^1 -action on $Z_{n,k,c}$ is free, and because c is regular, Marsden–Weinstein reduction gives a smooth symplectic manifold $\mathcal{R}_{n,k,c}$. This proves (i).

The circle actions τ and ρ commute, so the finite subgroup $C_k \subset S^1$ preserves $Z_{n,k,c}$ and descends to a symplectic action on $\mathcal{R}_{n,k,c}$. Quotients of symplectic manifolds by finite symplectic group actions are symplectic orbifolds; concretely, for $x \in \mathcal{R}_{n,k,c}$ one chooses a C_k -invariant compatible almost complex structure and metric, takes an equivariant exponential chart at x , and linearizes the stabilizer action. Equivariant Darboux coordinates

then identify the symplectic form with its tangent value, giving the local linear symplectic quotient model [13]. The action is effective: for $k \geq 2$, points with both Y_0 and Y_1 nonzero are dense on each nonempty level, since one can perturb the order vector while keeping its norm fixed. At such a point, $\rho^r(X, Y) = \tau_\alpha(X, Y)$ first forces $e^{i\alpha} = 1$ from Y_0 , and then $r = 0$ in C_k from Y_1 . For $k = 1$ the group is trivial. This proves (ii).

For (iii), average a Riemannian metric over H and use the equivariant exponential map to identify the fixed locus locally with $(T_x \mathcal{R}_{n,k,c})^H$. If $v \neq 0$ is fixed, choose w with $\omega(v, w) \neq 0$ and average w over H . Its average is fixed and has the same nonzero pairing with v , proving nondegeneracy on the fixed tangent subspace. Only finitely many subgroups occur, so the stated orbit-type subset is open in the fixed locus. The ambient quotient chart is $T_x \mathcal{R}_{n,k,c}/H$; the stratum corresponds to fixed tangent directions, not the entire quotient chart. $\square$

Corollary A.10. *For an embedded pattern p , the residual stabilizer after transposition reduction is*

$$\{r \in \mathbb{Z}/k\mathbb{Z} : \delta_p(m+r) = \delta_p(m) \text{ for every } m\}.$$

If the least positive gap period is $d \mid k$, its order is k/d . Trivial isotropy occurs exactly for full gap period k .

Proof. The stabilizer condition is $\rho^r(X, Y) = \tau_\alpha(X, Y)$ for some α . Inverse order DFT makes this $\eta_n^{p_{m+r}} = e^{i\alpha} \eta_n^{p_m}$. At $m = 0$ the phase is an n th root, so this is equivalent to $p_{m+r} = p_m + t$ for a constant t . Taking differences gives gap periodicity. Conversely periodic gaps make that difference constant by telescoping; the resulting transposition preserves the content and gives the content-coordinate equality as well. $\square$

For $(0, 4, 8)$ in G_{12} , the only nonzero retained content modes are $X_3 = X_6 = X_9 = 3$ and $Y_1 = 3$. Thus $\rho(X, Y) = \tau_{2\pi/3}(X, Y)$ and the reduced stabilizer is all C_3 , although the unreduced stabilizer is trivial. For a general point with $Y_{\ell_0} \neq 0$, the reduced stabilizer consists of r satisfying $k \mid r(\ell - \ell_0)$ for every supported order mode and $k \mid rj\ell_0$ for every supported content mode. Its order is the gcd of k and these differences/products.

Proposition A.11. *The inversion involution $\mathcal{I}$ descends to an anti-symplectic involution*

$$\bar{\mathcal{I}} : \mathcal{Q}_{n,k,c} \rightarrow \mathcal{Q}_{n,k,c}.$$

On every smooth orbifold stratum of $\mathcal{Q}_{n,k,c}$, the fixed-point set of $\bar{\mathcal{I}}$, when nonempty, is Lagrangian.

Proof. Because $I_j \circ \mathcal{I} = I_j$ and $J_\ell \circ \mathcal{I} = J_{-\ell}$, it preserves H_T and therefore $Z_{n,k,c}$. The relation

$$\mathcal{I} \circ \tau_\alpha = \tau_{-\alpha} \circ \mathcal{I}$$

shows that $\mathcal{I}$ descends through symplectic reduction to an involution of $\mathcal{R}_{n,k,c}$. Since $\mathcal{I}$ commutes with the finite C_k -action,

$$\mathcal{I} \circ \rho_{2\pi r/k} = \rho_{2\pi r/k} \circ \mathcal{I},$$

it descends to $\mathcal{Q}_{n,k,c} = \mathcal{R}_{n,k,c}/C_k$ and defines an involution $\bar{\mathcal{I}}$ on $\mathcal{Q}_{n,k,c}$. Since [theorem 8.6](#) gives $\mathcal{I}^*\omega = -\omega$, the descended map is anti-symplectic on every orbifold chart.

Commutation with C_k preserves residual orbit types. On each smooth orbit-type stratum apply the fixed-involution argument directly: average a Riemannian metric under the involution, use its exponential chart, and split the tangent space into the positive and negative eigenspaces. Anti-symplecticity makes each isotropic, and nondegeneracy pairs them, so both have half the stratum dimension. Each connected fixed component is therefore Lagrangian within that stratum. No single smooth branch across singular strata is asserted. $\square$

APPENDIX B. ALGORITHMIC EXTRACTION OF THE LAYERED COORDINATES

For completeness we record the exact computational recipe for a pattern $p = (p_0, \dots, p_{k-1}) \in \mathcal{P}_{n,k}$.

Step 1: content coordinates. Set $S = \{p_0, \dots, p_{k-1}\}$ and compute

$$X_j = \sum_{x \in S} \eta_n^{-jx}, \quad 1 \leq j \leq n-1.$$

Step 2: order coordinates. Compute the order signal

$$s_p(m) = \eta_n^{p_m}, \quad 0 \leq m \leq k-1,$$

and then

$$Y_\ell = \sum_{m=0}^{k-1} s_p(m) \zeta_k^{-\ell m}, \quad 0 \leq \ell \leq k-1.$$

Step 3: gap-spectrum coordinates (optional but musically informative). Compute the gap word

$$\delta_m = p_{m+1} - p_m \pmod{n},$$

the gap signal

$$g_p(m) = \eta_n^{\delta_m},$$

and its transform

$$G_\ell = \sum_{m=0}^{k-1} g_p(m) \zeta_k^{-\ell m}.$$

The support of G detects interval-cycle periodicity by [theorem 4.1](#). These G_ℓ are auxiliary interval-cycle coordinates in the present paper. Adding an independent G -factor with a standard symplectic form would make its squared-modulus actions $|G_\ell|^2/2$ moment-map coordinates.

Step 4: ambient and polar coordinates. The preceding steps give the embedded seed point $\iota_{n,k}(p)$ in the ambient space with the globally defined form (7.1). Set $I_j = |X_j|^2/2$ and $J_\ell = |Y_\ell|^2/2$ for all modes. For a nonzero coefficient one may also write

$$X_j = \sqrt{2I_j}e^{i\theta_j}, \quad Y_\ell = \sqrt{2J_\ell}e^{i\phi_\ell}.$$

The full action-angle expression (7.2) applies only when every coefficient is nonzero. At zero coefficients retain Cartesian coordinates; no phase angle is defined there. For example, the ascending full aggregate has all retained $X_j = 0$ and only $Y_1 = n$ nonzero. It is a valid seed even though it lies outside the full action-angle locus.

APPENDIX C. ON DELIBERATE REDUNDANCY

The order signal already determines the whole ordered pattern, whereas the content signal by itself remembers only the unordered content. Accordingly, the layered embedding is redundant. This redundancy is deliberate: it separates the two musically meaningful notions of structure. The content coordinates support content autocorrelation and transposition/inversion analysis; the order coordinates support cyclic-order analysis; and the symplectic form splits into corresponding layers. The redundancy is therefore not a defect but a structural refinement.

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